

# Shriyansh Sinha

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## EDUCATION

**Indian Institute of Technology Patna** — Patna, India

Nov 2021 – May 2025

B.Tech | GPA: 8.0 / 10

## EXPERIENCE

**Software Engineer I** | Juspay

May 2025 – Present

Python, Haskell, Rescript, AWS, kubernetes, Docker, PostgreSQL, kibana

- Architected distributed backend service handling 50K+ daily transactions with sub-100ms p99 latency, owning full lifecycle from design through production deployment
- Built a generative AI summarization pipeline on AWS Bedrock integrating LLM inference over 5K+ daily code events; automated alerting and actionable digest generation eliminated 10+ hours/week of manual engineering reporting.
- Owned CI/CD release pipelines for 5+ services, implementing blue-green deployments and automated rollback triggers that eliminated deployment-related incidents
- Engineered performance management system aggregating metrics across 2000+ engineers, auto-generating quarterly review reports and saving 15+ manager-hours per cycle

**Software Engineering Intern** | Juspay

Dec 2024 – May 2025

Go, Haskell, NixOS, Grafana, Jenkins, Kafka, ClickHouse, Azure

- Designed fault-tolerant data pipelines processing 2M+ daily events through Kafka into ClickHouse, achieving 99.9% delivery guarantee with <500ms end-to-end latency
- Built customer analytics sessionization service from merchant requirements to production, enabling real-time user journey tracking across payment flows
- Streamlined CI/CD workflows in Jenkins, reducing deployment time by 30% and adding automated rollback on failures.

## RESEARCH & PUBLICATIONS

**HyDeML: Hybrid Mode Decomposition-based Machine Learning Framework for Bus Arrival Prediction**

[[Paper](#)]

Washington D.C., 2026 105th Transportation Research Board Annual Meeting

- Built and benchmarked 20+ ML pipelines combining 3 signal decompositions (EMD, EEMD, VMD)  $\times$  5 architectures (SVM, ANN, GD, LSTM, Transformers); achieved 40% MAPE reduction, outperforming LSTM and Transformer baselines
- Processed 1.2K+ GPS observations per segment across 70 road sections (28km route); optimized inference to <1s per trip for real-time deployment viability

## PROJECTS

**Citation-Grounded RAG QA System** — Generative AI for Document Intelligence

[[Github Link](#)] [[Demo](#)]

Django, React, ChromaDB, HuggingFace, Ollama, LangChain

- Developed RAG-based document QA system over PDFs with citation-backed answers (document name + page number) and session-isolated storage supporting concurrent multi-user querying
- Built semantic search pipeline (embeddings  $\rightarrow$  vector DB  $\rightarrow$  LLM reranking) achieving high-precision retrieval with hallucination-resistant responses

**AutoDeploy** — Automated Deployment Platform

[[Github Link](#)] [[Demo](#)]

JavaScript, C++, Redis, Azure DB, Simple-Git

- Built one-click deployment platform from Git URL to live hosting in <20 seconds; architected distributed 3-tier system with Redis queue orchestration and autoscaling workers
- Eliminated manual DevOps overhead for small teams with isolated upload, deployment, and serving layers enabling independent horizontal scaling

## SKILLS

**Languages:** Java, Python, Haskell, C/C++, JavaScript, SQL, TypeScript

**Infrastructure:** AWS, Azure, Docker, Kafka, Redis, PostgreSQL, ClickHouse, Cassandra, Kubernetes

**AI/ML & Monitoring:** LLMs, RAG, Vector DBs, PyTorch, MCP servers, Time-Series Forecasting, Grafana, Kibana, Prometheus

**Systems:** Microservices, CI/CD, Distributed Systems, Event-Driven Architecture, Linux

## COMPETITIVE PROGRAMMING

- Codeforces Expert (1767) • LeetCode Knight (1883) • CodeChef 4-Star (1835) • Global Rank 116 in Codenite 2025

## LEADERSHIP

**Cultural Secretary**, IIT Patna Students' Body (2023–24) — led 50+ events for 10,000+ students

**Web Development Lead**, Anwesha - Cultural Fest — managed 8-person team delivering impact to 30k participants